I Can turn an array to multiply

Level 3
Operations
Teacher 1

CCSS 3.OA.B.5 · **EE** none · access point M.EE.N.CN.2 · **CRA:** Concrete → Pictorial · **Flow:** Standard (3 new routine steps)

Lesson guidance:

Tip: Tray first, then the picture. Turn a real tray of counters before any drawn array.

Tip: No square in the I Do. A square turns into itself. It waits for the last row.

Watch: Recounting. A pupil who recounts the turned array has not got step 5 yet.

Lesson flow:

#	Section	Pupil page	Min
1	Prerequisite Check	Prerequisite Check	6
2	Number sense	Number sense	5
3	Spaced retrieval	Mini whiteboards	4
4	Chaining bridge	Mini whiteboards	2
5	I Do: the anchor chart	Anchor chart	10
6	We Do: Guided Practice	Lesson sheet	8
7	Check Question	Mini whiteboards	3
8	You Do: Independent Practice	Practice 1–2	9
9	Word problem	Word problem	5
10	Plenary and exit slip	Exit slip	4
	Transitions	between pages	4
	Total		**60**

Materials:

- ☐ 12 counters on a tray or card, one per pair
- ☐ Dot paper for the board
- ☐ Pupil packet, one per pupil
- ☐ Mini whiteboards and pens
- ☐ Anchor chart
- ☐ Exit slips, cut in half

Vocabulary:

• • • / • • •	• • • •	2 × 3 ↻ 3 × 2
array	**row**	**turn**
dots in equal rows	a line of dots across	move it a quarter way round

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I Can turn an array to multiply

Level 3
Operations
Teacher 2

What students need to know:

#	Skill or Concept	Atom Type	Teacher Notes
1	Count by 2, 5, 10	Fact · Pre-requisite	Prerequisite Check a.
2	× means "groups of", "rows of"	Fact · Pre-requisite	Prerequisite Check b–e.
3	**Array fact:** rows × in each row	Conceptual · Pre-requisite	Prerequisite Check f–g.
4	**Turned array:** rows become columns	Comparative	Non-example: a dot added.
5	**Same product: turning adds no dot**	**Conceptual**	**Critical Atom. Ask "Did a dot move away?"**
6	Known fact gives the turned fact	Conceptual	Use the fact. Do not recount.

What students do:

Step	Task / Routine atom	Tag	Check it with
1	Look at the array: count the rows and in each row.	Pre-requisite	"How many rows?"
2	Write the fact.	Pre-requisite	Guided cell 1
3	Turn the array: rows become columns.	New	Tray: "How many rows now?"
4	Write the new fact.	New	Guided cell 2
5	**Check: same product. No recount.**	**New**	**Check Question; exit b**

Rows tagged New: 3 → Flow: Standard. One chunk: turn the array; the product stays the same.

How to teach:

Turn a real tray of counters first: pupils count once, then see the same total. Ask "Did a dot move away?" First examples: non-square, no repeated digit.

Board: Dot paper: 4 rows of 2, then turned.

Common mistakes:

Mistake	Step	Looks like	Fix	Caught by
Recounts: a new product	5	8 × 4 = 28	"Did a dot move away?"	Check Question A, Exit b
Adds the two factors	5	8 × 4 = 12	"Times, not plus: same product."	Check Question B
Copies the fact: no swap	4	2, 3, 6 for 3 × 2 = 6	Count the turned rows aloud.	Exit a · Practice 1
Product as a factor	4	36 in the box of exit c	The product is always last.	Exit c · Practice 2

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I Can turn an array to multiply

Level 3
Operations
Reference

Reference key:

Pages and order

Page	Tab	Score
P0	Prerequisite	one per question
P1	Number sense	/10
P2	Anchor chart	none (a poster)
P3	Lesson 1	Independent cells
P4	Practice n / Mixed n	every cell
P5	Word problem	/1
P6	Exit slip	/3 a half
T1–T5	Teacher n	—
T6	Answer Key	—

Fade levels

Label	Means
★	Guided: hints on, step 1 traced
★★	Independent: hints off
★★★	Last row: the edge cases

Data loop

● **Most:**	Go on.
◑ **About half:**	Short reteach, then go on.
○ **Fewer than half:**	Stop. Reteach with a concrete example.

Abbreviations

Short	In full
MWBs	mini whiteboards
ELL	English language learners
SPED	special education students
RCDWS	Read, Circle, Draw, Write, Solve
CRA	Concrete-Representational-Abstract

Routing ids

No site lesson exists yet for these prerequisites: T6 names a CCSS code and a short title (V22 Appendix H).

Answer slots

Shape	Used for
_____	a number after =
☐	a part inside an expression
☐	a tick (routine panel, exit slip)

Icons

Icon	Means
	Tip
	Feedback
	Watch
	Research
	Critical Atom
	Language tip
	Why it works
	Board
1 2 3 4 5 ☑	The pupil Steps (accent)

Band labels

Pupil: Vocabulary · Remember · Guided Practice · Steps · Independent Practice · Mixed Review · Number sense: A / B · Workspace · Another example · Say · Rule.

Teacher: Lesson guidance · Lesson flow · Materials · What students need to know · What students do · How to teach · Common mistakes · Section n · Prerequisite routing · Accepted answers · Answers at a glance.

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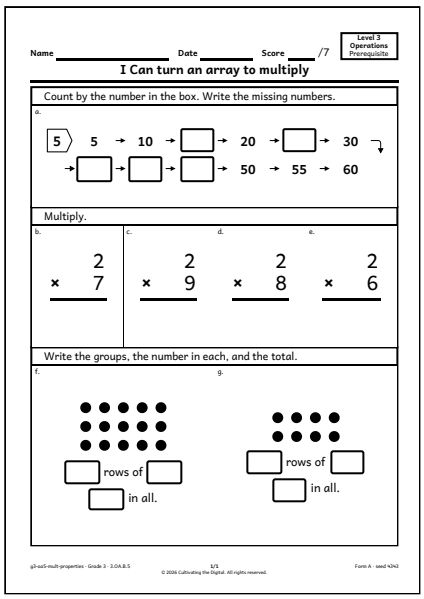
I Can turn an array to multiply

Level 3 Operations Teacher 3

Section 1 · Prerequisite Check (6 min)

Tip: Route before teaching. Mark it while pupils do the Number sense page.

1. **Do:** Hand out the Prerequisite Check, face down.
2. **Say:** "Do these on your own. Try every one."
3. **Do:** Walk and mark as pupils finish.
   Note: Expected: **a. 15, 25, 35, 40, 45 · b. 14 · c. 18 · d. 16 · e. 12 · f. 15 · g. 8**. Routing is on the last page (T6).



Name _____ Date _____ Score ___ /7 — Level 3 Operations Prerequisite

I Can turn an array to multiply

Count by the number in the box. Write the missing numbers.

a. 5 → 5 → 10 → ☐ → 20 → ☐ → 30 → ☐ → ☐ → ☐ → 50 → 55 → 60

Multiply.

b. 2 × 7 = ☐

c. 2 × 9 = ☐

d. 2 × 8 = ☐

e. 2 × 6 = ☐

Write the groups, the number in each, and the total.

f. ☐ rows of ☐ ☐ in all.

g. ☐ rows of ☐ ☐ in all.

g3-oa5-mult-properties · Grade 3 · 3.OA.B.5 — 1/1 — Form A · seed 4242

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Section 2 · Number sense (5 min)

1. **Do:** Hand out the Number sense page.
2. **Say:** "Band A: read each row. Write the answer."
3. **Ask:** "What is 10 more than 47?"
   Note: Expected: **57**. Band B: the same five kinds, harder numbers.

Section 3 · Spaced retrieval (4 min)

1. **Do:** Pupils hold mini whiteboards. Write each fact on the board.
2. **Ask:** "5 × 5 = ?"
   Note: Expected: **25**.
3. **Ask:** "2 × 10 = ?"
   Note: Expected: **20**.
4. **Ask:** "10 × 5 = ?"
   Note: Expected: **50**.

Section 4 · Chaining bridge (2 min)

1. **Say:** "Last time: one array gives a fact."
2. **Say:** "Today: one array gives two facts. Same answer."
3. **Do:** Show the tray. Pupils count the counters once.
   Note: They keep the total in mind for the turn.

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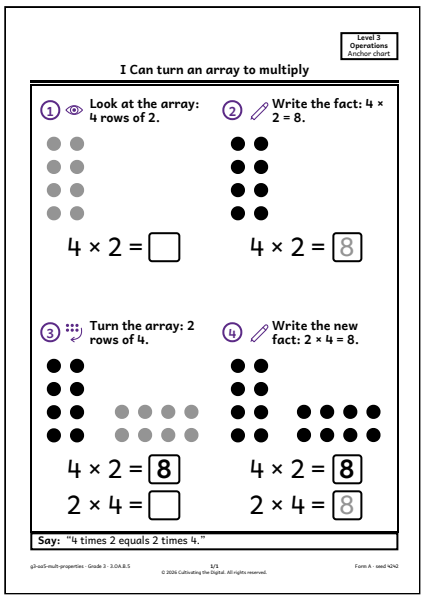
I Can turn an array to multiply

Level 3 Operations Teacher 4

Section 5 · I Do: the anchor chart (10 min)

Watch: Turn slowly. Let them see that no dot leaves the tray.

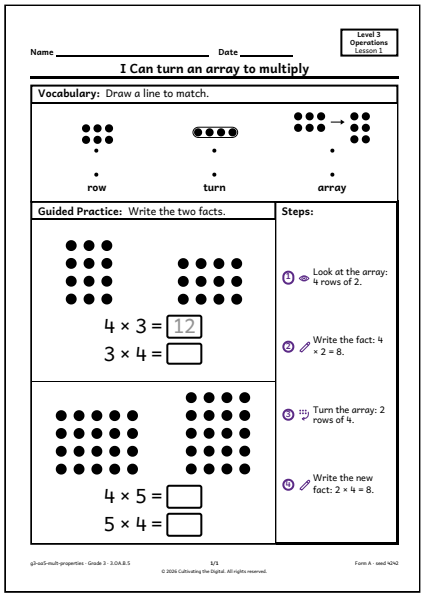
1	**Do:**	Hold up a tray of 4 rows of 2.
2	**Say:**	"4 rows of 2 is 8."
	Note:	Step 1 and 2: write **4 × 2 = 8**.
3	**Do:**	Turn the tray a quarter way round. Slowly.
	Note:	Step 3.
4	**Ask:**	"How many rows now? How many in each row?"
	Note:	Expected: **2** rows of **4**.
5	**Say:**	"2 rows of 4 is 8."
	Note:	Step 4: write **2 × 4 = 8**.
6	**Ask:**	"Did a dot move away?"
	Note:	Expected: no. Step 5: same product.
7	**Say:**	"4 times 2 equals 2 times 4."
	Note:	The Say frame. Pupils repeat twice.
8	**Do:**	Cover the dots. Point to 4 × 2 = 8 on the chart.
	Note:	One new thing: no dots. Practice 2 and exit b, c.
9	**Ask:**	"So what is 2 × 4?"
	Note:	Expected: **8**. Same two numbers, same product.
10	**Do:**	Write 2 × □ = 8 on the board.
	Note:	Expected: **4**: the factor that moved.



Section 6 · We Do: Guided Practice (8 min)

Feedback: Fix it at once. Fix: turn the tray again and ask "Same dots?"

1	**Say:**	"Look at the first array. Count the rows."
2	**Ask:**	"How many rows? How many in each row?"
	Note:	Expected ★: **4 × 3 = 12**; turned **3 × 4 = 12**.
3	**Do:**	Pupils do the second array on their own.
	Note:	Expected ★: **4 × 5 = 20**; turned **5 × 4 = 20**.
4	**Ask:**	"Turn it. What is the new fact?"
	Note:	Pupils say it; write it on the board.
5	**Say:**	"Cover the dots of the second array. Read its first fact."
	Note:	Expected: **4 × 5 = 20**; no dots, the same turned fact.



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I Can turn an array to multiply

Level 3 Operations Teacher 5

Section 7 · Check Question (3 min)

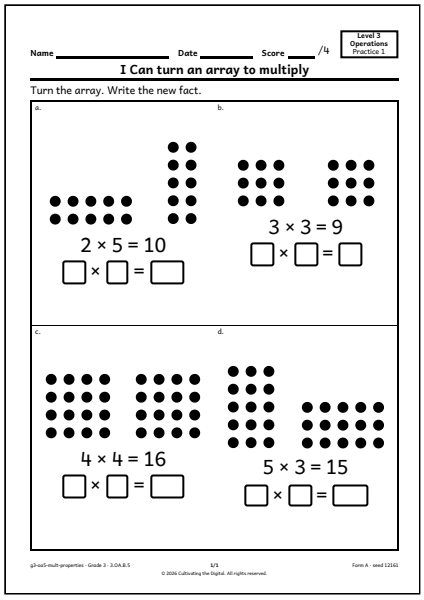
1. **Do:** Write 4 × 8 = 32 on the board. No dots.
2. **Do:** Write three turned facts. A: 8 × 4 = 28 · B: 8 × 4 = 12 · C: 8 × 4 = 32.
3. **Ask:** "Which turned fact is true? Show fingers."
   Note: Correct: **C 8 × 4 = 32**. A: Recounts: a new product · B: Adds the two factors

- ● **Most:** Go on to You Do.
- ◑ **About half:** Reteach step 5 with the tray, then two more.
- ○ **Fewer than half:** Stop. Whole class: We Do again with a new tray.

Section 8 · You Do: Independent Practice (9 min)

Watch: The last row. Walk to the last row first: Practice 1: 16 (a square), 15 (a 5); Practice 2: 49 (a square), 35 (a 5).

1. **Do:** Pupils finish the lesson sheet, then Practice 1 and 2.
2. **Say:** "Write the fact. Then say the turned fact."
3. **Do:** Circulate at ★★. Look for pupils who recount.
   Note: Ask "Did a dot move away?"


Name _____ Date _____ Score ___/4 Level 3 Operations Practice 1

I Can turn an array to multiply

Turn the array. Write the new fact.

a. 2 × 5 = 10 ☐ × ☐ = ☐

b. 3 × 3 = 9 ☐ × ☐ = ☐

c. 4 × 4 = 16 ☐ × ☐ = ☐

d. 5 × 3 = 15 ☐ × ☐ = ☐

g3-oa5-mult-properties · Grade 3 · 3.OA.B.5 1/1 Form A · seed 12161

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Section 9 · Word problem (5 min)

1. **Do:** Read the story aloud two times. Pupils tick each box.
2. **Ask:** "How many rows? How many in each row?"
   Note: Expected: **6** rows, **7** in each.
3. **Ask:** "How many eggs? Say both facts."
   Note: Expected: **42**; said: 6 × 7 = 42 and 7 × 6 = 42.

Section 10 · Plenary and exit slip (4 min)

1. **Say:** "Turning does not change the total."
   Note: Say it together twice.
2. **Do:** Hand out the exit slip. No help. Then sort the slips.
   Note: a. **3, 2, 6** · b. **27** · c. **6**.

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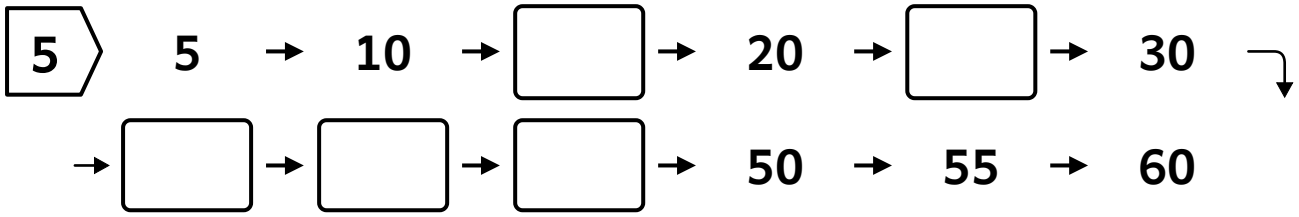
Name _____ **Date** _____ **Score** _____ /7

Level 3
Operations
Prerequisite

I Can turn an array to multiply

Count by the number in the box. Write the missing numbers.

a.



Multiply.

b.

$$\begin{array}{r} 2 \\ \times\ 7 \\ \hline \end{array}$$

c.

$$\begin{array}{r} 2 \\ \times\ 9 \\ \hline \end{array}$$

d.

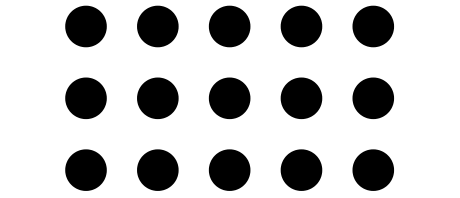
$$\begin{array}{r} 2 \\ \times\ 8 \\ \hline \end{array}$$

e.

$$\begin{array}{r} 2 \\ \times\ 6 \\ \hline \end{array}$$

Write the groups, the number in each, and the total.

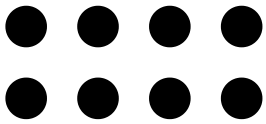
f.



_____ rows of _____

_____ in all.

g.



_____ rows of _____

_____ in all.

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Name _____ Date _____ Score _____ /10

Level 3
Operations
Number sense

I Can turn an array to multiply

Number sense: A

1. Write the missing number. **10** more than **47** is _____ .
2. Write the missing number. **10** less than **20** is _____ .
3. Write <, > or = in the circle. **72** ◯ **35**
4. Write how many of each place. **39** = ☐ tens ☐ ones
5. Write the missing number. **1** more than **33** is _____ .

Number sense: B

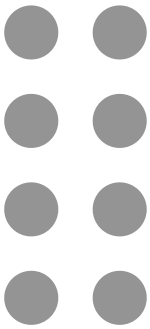
6. Write the missing number. **10** more than **88** is _____ .
7. Write the missing number. **10** less than **93** is _____ .
8. Write <, > or = in the circle. **54** ◯ **56**
9. Write how many of each place. **84** = ☐ tens ☐ ones
10. Write the missing number. **1** more than **59** is _____ .

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Level 3
Operations
Anchor chart

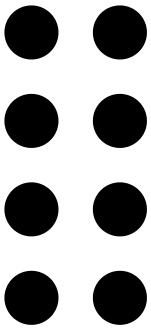
I Can turn an array to multiply

1 **Look at the array: 4 rows of 2.**



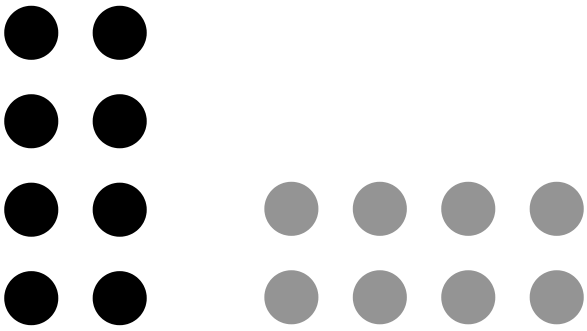
$4 \times 2 = \square$

2 **Write the fact: 4 × 2 = 8.**



$4 \times 2 = 8$

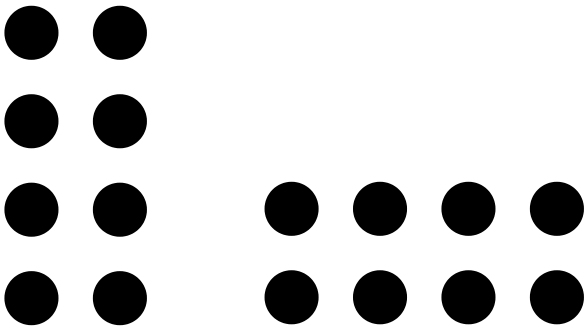
3 **Turn the array: 2 rows of 4.**



$4 \times 2 = 8$

$2 \times 4 = \square$

4 **Write the new fact: 2 × 4 = 8.**



$4 \times 2 = 8$

$2 \times 4 = 8$

Say: "4 times 2 equals 2 times 4."

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Name _____ Date _____

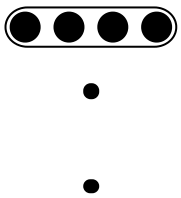
Level 3
Operations
Lesson 1

I Can turn an array to multiply

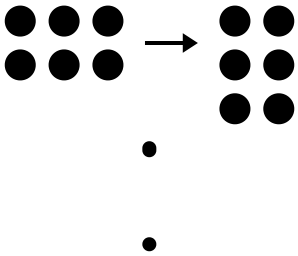
Vocabulary: Draw a line to match.



row

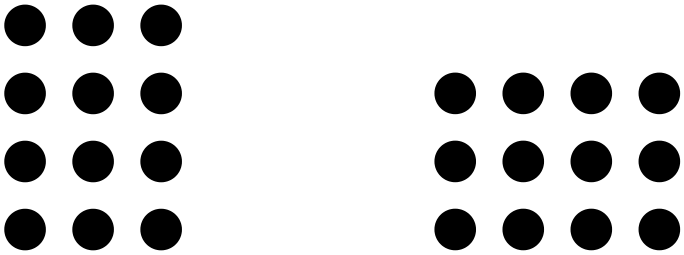


turn



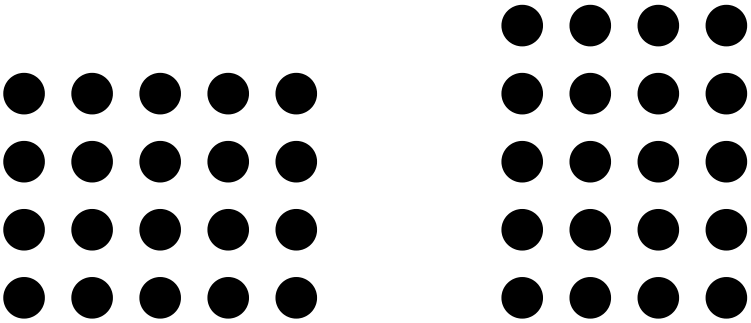
array

Guided Practice: Write the two facts.



$4 \times 3 = 12$

$3 \times 4 = \square$



$4 \times 5 = \square$

$5 \times 4 = \square$

Steps:

1. Look at the array: 4 rows of 2.
2. Write the fact: 4 × 2 = 8.
3. Turn the array: 2 rows of 4.
4. Write the new fact: 2 × 4 = 8.

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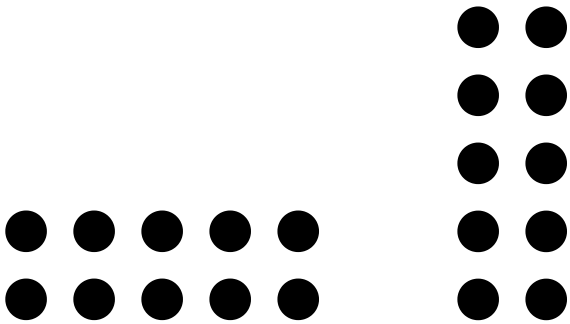
Name _____ Date _____ Score _____ /4

Level 3 Operations Practice 1

I Can turn an array to multiply

Turn the array. Write the new fact.

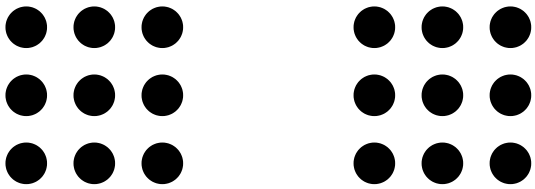
a.



$2 \times 5 = 10$

$\square \times \square = \square$

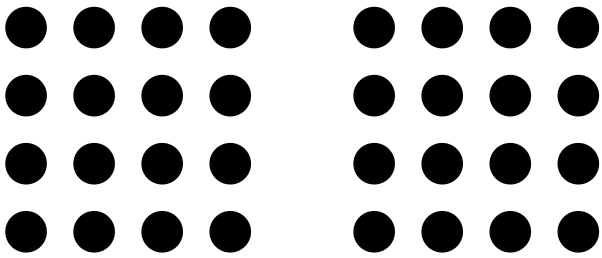
b.



$3 \times 3 = 9$

$\square \times \square = \square$

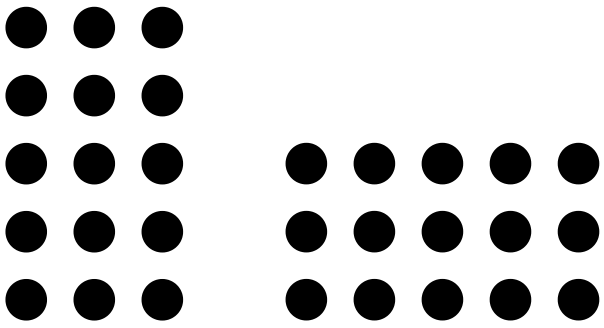
c.



$4 \times 4 = 16$

$\square \times \square = \square$

d.



$5 \times 3 = 15$

$\square \times \square = \square$

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Name _____ Date _____ Score _____ /10

Level 3 Operations Practice 2

I Can turn an array to multiply

Write the missing number.

a.

$2 \times 8 = 16$

$8 \times 2 = \square$

b.

$8 \times 3 = 24$

$3 \times 8 = \square$

c.

$9 \times 2 = 18$

$2 \times \square = 18$

d.

$4 \times 9 = 36$

$9 \times 4 = \square$

e.

$9 \times 6 = 54$

$6 \times 9 = \square$

f.

$4 \times 7 = 28$

$7 \times 4 = \square$

g.

$2 \times 6 = 12$

$6 \times \square = 12$

h.

$6 \times 8 = 48$

$8 \times 6 = \square$

i.

$7 \times 7 = 49$

$7 \times 7 = \square$

j.

$7 \times 5 = 35$

$5 \times 7 = \square$

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Name _____ Date _____ Score _____ /1

Level 3
Operations
Word problem

I Can turn an array to multiply

Read the story. Draw it. Solve.

A box has 6 rows of eggs.

There are 7 eggs in each row.

How many eggs are in the box?

There are _____ eggs.

- ☐ Read it two times.
- ☐ Circle the numbers.
- ☐ Underline the question.
- ☐ Draw it.
- ☐ Write the equation.
- ☐ Solve. Check.

Workspace:

☐ × ☐ = ☐

Say: "There are _____ eggs."

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Name _____ Score _____ /3

Level 3 Operations Exit slip

I Can turn an array to multiply

Turn the array. Write the new fact.

a.



$2 \times 3 = 6$

$\square \times \square = \square$

Write the missing number.

b.

$3 \times 9 = 27$

$9 \times 3 = \square$

c.

$6 \times 6 = 36$

$6 \times \square = 36$

Say: "3 times 9 equals 9 times 3."

☐ I can do this. ☐ I need one more day.

Name _____ Score _____ /3

Level 3 Operations Exit slip

I Can turn an array to multiply

Turn the array. Write the new fact.

a.



$2 \times 3 = 6$

$\square \times \square = \square$

Write the missing number.

b.

$3 \times 9 = 27$

$9 \times 3 = \square$

c.

$6 \times 6 = 36$

$6 \times \square = 36$

Say: "3 times 9 equals 9 times 3."

☐ I can do this. ☐ I need one more day.

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Answer Key _____ Date _____ Score _____ /7

Level 3 Operations Answer Key

I Can turn an array to multiply

Count by the number in the box. Write the missing numbers.

a.

5 → 5 → 10 → 15 → 20 → 25 → 30 →

35 → 40 → 45 → 50 → 55 → 60

Multiply.

b.

$$\begin{array}{r} 2 \\ \times\ 7 \\ \hline 14 \end{array}$$

c.

$$\begin{array}{r} 2 \\ \times\ 9 \\ \hline 18 \end{array}$$

d.

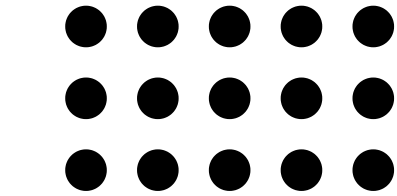
$$\begin{array}{r} 2 \\ \times\ 8 \\ \hline 16 \end{array}$$

e.

$$\begin{array}{r} 2 \\ \times\ 6 \\ \hline 12 \end{array}$$

Write the groups, the number in each, and the total.

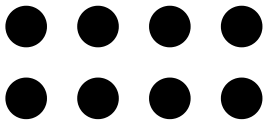
f.



3 rows of 5

15 in all.

g.



2 rows of 4

8 in all.

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Answer Key _____ Date _____ Score _____ /10

Level 3 Operations Answer Key

I Can turn an array to multiply

Number sense: A

1. Write the missing number. **10** more than **47** is **57**.
2. Write the missing number. **10** less than **20** is **10**.
3. Write <, > or = in the circle. **72** **>** **35**
4. Write how many of each place. **39** = **3** tens **9** ones
5. Write the missing number. **1** more than **33** is **34**.

Number sense: B

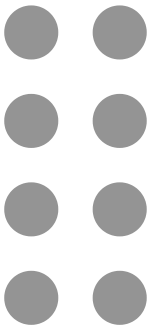
6. Write the missing number. **10** more than **88** is **98**.
7. Write the missing number. **10** less than **93** is **83**.
8. Write <, > or = in the circle. **54** **<** **56**
9. Write how many of each place. **84** = **8** tens **4** ones
10. Write the missing number. **1** more than **59** is **60**.

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Level 3
Operations
Answer Key

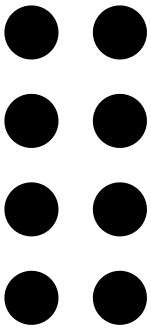
I Can turn an array to multiply

1 **Look at the array: 4 rows of 2.**



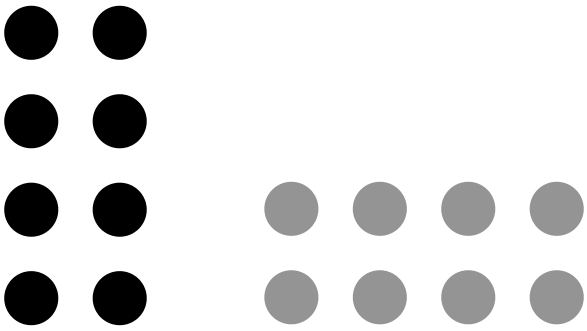
$4 \times 2 =$ ☐

2 **Write the fact: 4 × 2 = 8.**



$4 \times 2 = 8$

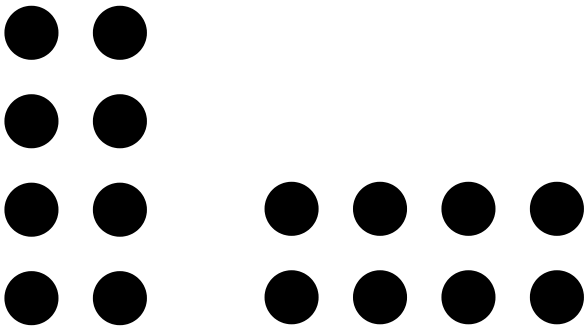
3 **Turn the array: 2 rows of 4.**



$4 \times 2 = 8$

$2 \times 4 =$ ☐

4 **Write the new fact: 2 × 4 = 8.**



$4 \times 2 = 8$

$2 \times 4 = 8$

Say: "4 times 2 equals 2 times 4."

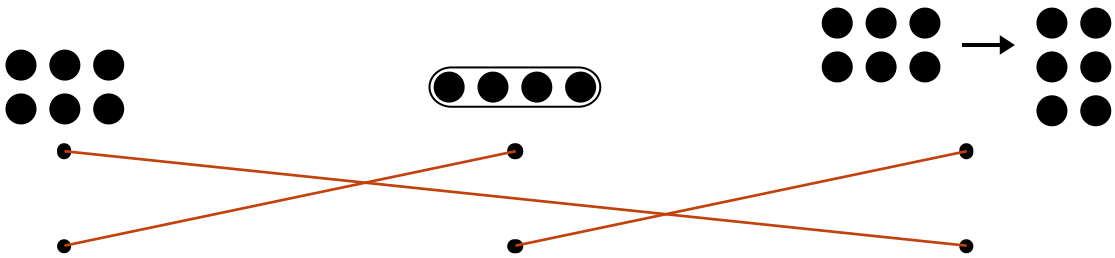
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Answer Key _____ Date _____

Level 3
Operations
Answer Key

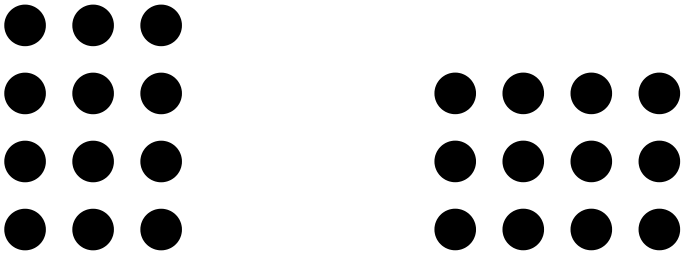
I Can turn an array to multiply

Vocabulary: Draw a line to match.



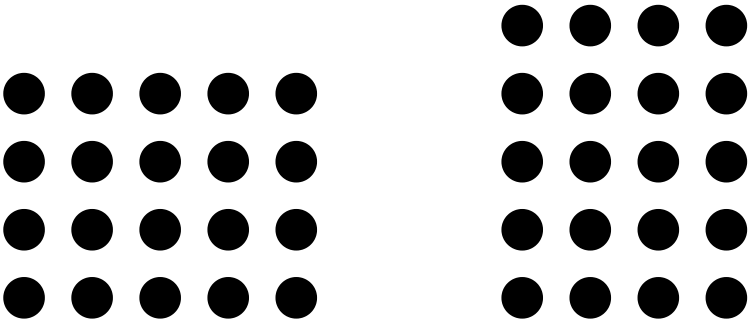
row turn array

Guided Practice: Write the two facts.



$4 \times 3 = \boxed{12}$

$3 \times 4 = \boxed{12}$



$4 \times 5 = \boxed{20}$

$5 \times 4 = \boxed{20}$

Steps:

1. Look at the array: 4 rows of 2.
2. Write the fact: 4 × 2 = 8.
3. Turn the array: 2 rows of 4.
4. Write the new fact: 2 × 4 = 8.

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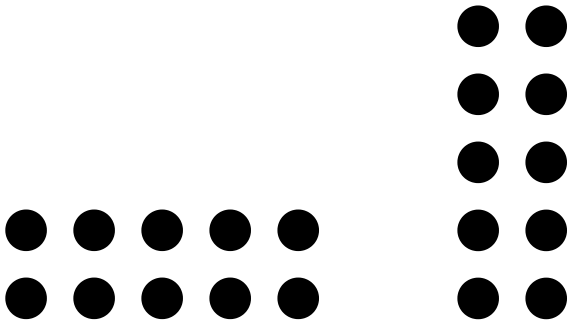
Answer Key _____ **Date** _____ **Score** _____ /4

Level 3 Operations Answer Key

I Can turn an array to multiply

Turn the array. Write the new fact.

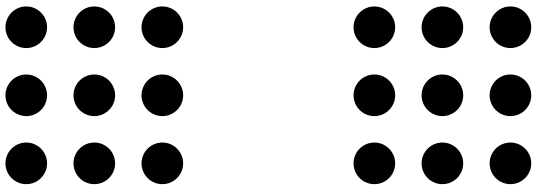
a.



$2 \times 5 = 10$

$5 \times 2 = 10$

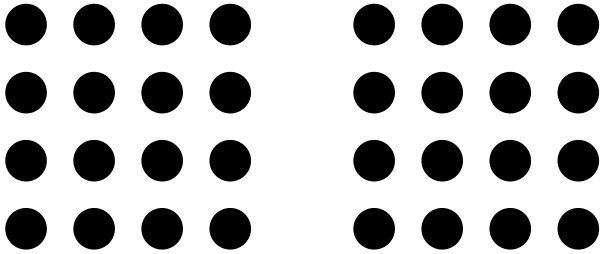
b.



$3 \times 3 = 9$

$3 \times 3 = 9$

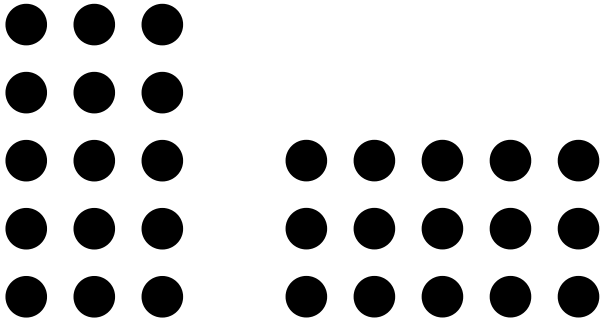
c.



$4 \times 4 = 16$

$4 \times 4 = 16$

d.



$5 \times 3 = 15$

$3 \times 5 = 15$

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Answer Key _____ **Date** _____ **Score** _____ /10

Level 3 Operations Answer Key

I Can turn an array to multiply

Write the missing number.

a.

$2 \times 8 = 16$

$8 \times 2 =$ **16**

b.

$8 \times 3 = 24$

$3 \times 8 =$ **24**

c.

$9 \times 2 = 18$

$2 \times$ **9** $= 18$

d.

$4 \times 9 = 36$

$9 \times 4 =$ **36**

e.

$9 \times 6 = 54$

$6 \times 9 =$ **54**

f.

$4 \times 7 = 28$

$7 \times 4 =$ **28**

g.

$2 \times 6 = 12$

$6 \times$ **2** $= 12$

h.

$6 \times 8 = 48$

$8 \times 6 =$ **48**

i.

$7 \times 7 = 49$

$7 \times 7 =$ **49**

j.

$7 \times 5 = 35$

$5 \times 7 =$ **35**

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Answer Key _____ **Date** _____ **Score** _____ /1

Level 3 Operations Answer Key

I Can turn an array to multiply

Read the story. Draw it. Solve.

A box has 6 rows of eggs.

There are 7 eggs in each row.

How many eggs are in the box?

There are **42** eggs.

- ☐ Read it two times.
- ☐ Circle the numbers.
- ☐ Underline the question.
- ☐ Draw it.
- ☐ Write the equation.
- ☐ Solve. Check.

Workspace:

6 × **7** = **42**

Say: "There are 42 eggs."

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Answer Key _____ Score _____ /3

Level 3 Operations Exit slip

I Can turn an array to multiply

Turn the array. Write the new fact.

a.



$2 \times 3 = 6$

$\boxed{3} \times \boxed{2} = \boxed{6}$

Write the missing number.

b. $3 \times 9 = 27$

$9 \times 3 = \boxed{27}$

c. $6 \times 6 = 36$

$6 \times \boxed{6} = 36$

Say: "3 times 9 equals 9 times 3."

☐ I can do this. ☐ I need one more day.

Answer Key _____ Score _____ /3

Level 3 Operations Exit slip

I Can turn an array to multiply

Turn the array. Write the new fact.

a.



$2 \times 3 = 6$

$\boxed{3} \times \boxed{2} = \boxed{6}$

Write the missing number.

b. $3 \times 9 = 27$

$9 \times 3 = \boxed{27}$

c. $6 \times 6 = 36$

$6 \times \boxed{6} = 36$

Say: "3 times 9 equals 9 times 3."

☐ I can do this. ☐ I need one more day.

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I Can turn an array to multiply

Level 3 Operations Answer Key

Prerequisite routing:

Question	Prerequisite (T2 atom)	If missed → Lesson
a. Count by 2, 5, 10	Fact, atom 1	2.NBT.A.2 Count by 5s and 10s
b–e. × means “groups of”, “rows of”	Fact, atom 2	3.OA.A.1 Equal groups
f–g. Array fact	Conceptual, atom 3	2.OA.C.4 Use arrays
	Rule	0 missed: go on · 1: that lesson · 2+: the earliest

Accepted answers:

- Prerequisite Check arrays: rows and in each row in either order.
- Exit a: only 3 × 2 = 6; writing 2 × 3 = 6 again is the no-swap mistake.
- Practice 2 and exit b, c: one number in each box, no working needed.
- Practice 1: a turned fact read off the turned array is right even if the pupil counted it.
- Egg story: 6 × 7 or 7 × 6; the label “eggs” may be left off.

Answers at a glance:

Page	Answers	Fade
P0 Prerequisite	**15, 25, 35, 40, 45 · 14 · 18 · 16 · 12 · 15 · 8**	★★
P1 Number sense	**57 · 10 · > · 3 tens 9 ones · 34 · 98 · 83 · < · 8 tens 4 ones · 60**	★★
P3 Guided	**12, 12 · 20, 20**	★
P4-1 Practice, rows 1–1	**5, 2, 10 · 3, 3, 9**	★★
P4-1 Practice, last row	**4, 4, 16 · 3, 5, 15**	★★★
P4-2 Practice, rows 1–4	**16 · 24 · 9 · 36 · 54 · 28 · 2 · 48**	★★
P4-2 Practice, last row	**49 · 35**	★★★
P5 Word problem	**6 × 7 = 42**	★
P6 Exit slip	**a. 3, 2, 6 · b. 27 · c. 6**	★★★

Wrong answers to look for when marking:

Pupil wrote	Likely mistake	Next step
24 (b)	**Recounts: a new product**	Tray: turn it; “Did a dot move away?”
2, 3, 6 (a)	**Copies the fact: no swap**	Count the turned rows aloud.
36 (c)	**Product as a factor**	The box map: the product is last.
3, 6, 6 (a)	**Product as a factor**	The product is always last.
12 (b)	**Adds the two factors**	“Times, not plus.” Say the fact aloud.
The first fact copied on Practice 1	**Copies the fact: no swap**	Turn the page a quarter way: count the rows.
A box left empty on Practice 2	**Not sure which number moved**	Point: the two factors swap places.

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